Minor Project Synopsis Report

THRYLOS

(ADMIN MANAGEMENT PORTAL)

Project Category: Industry

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1.ABSTRACT

Modern esports organizations and technology service companies operate in highly dynamic environments that demand efficient internal coordination, secure access control, and real-time operational visibility. Thrylos is an esports tournament hosting organization that also functions as a technology solutions provider, managing multiple operational domains such as employee workflows, technical development, social media engagement, and administrative governance. As the organization scales, manual tracking systems and disconnected tools become inadequate, leading to inefficiencies, data inconsistencies, and limited decision-making capability.

This project presents the design and development of a **Role-Based Admin Dashboard and Management System** tailored specifically for Thrylos. The proposed system centralizes organizational operations into a single web-based platform, enabling secure access through Role-Based Access Control (RBAC). Users are authenticated and granted permissions dynamically based on their assigned roles, such as Super Admin, Admin, Employee, Technical Team, and Social Media Team. Each role is provided with a customized dashboard and restricted access to relevant modules, ensuring both security and operational clarity.

The system integrates multiple functional modules including daily task management, attendance tracking, activity logging, employee management, social media analytics, technical work dashboards, audit logs, notification systems, bulk data upload, and file/media management. Real-time analytics and summary dashboards provide actionable insights into organizational performance, employee productivity, and engagement metrics. For example, administrators can monitor attendance trends, task completion rates, social media growth statistics, and system usage patterns from a unified interface.

From a technical perspective, the system is implemented using a modern web technology stack, ensuring scalability, maintainability, and responsiveness. The modular architecture allows independent enhancement of features while maintaining overall system integrity. Security considerations such as access control enforcement, audit logging, and data validation are incorporated to support enterprise-level reliability.

In conclusion, the Thrylos Role-Based Admin Dashboard addresses critical challenges in organizational management by providing a secure, scalable, and data-driven solution. The system enhances transparency, reduces administrative overhead, and supports informed decision-making, making it a practical and effective management platform for esports and technology-driven organizations.

2.INTRODUCTION

2.1 Background and Context

The rapid digital transformation of organizations has significantly altered how internal operations, workforce management, and decision-making processes are conducted. Esports organizations and technology service providers, in particular, operate within fast-paced, data-intensive environments where multiple teams collaborate simultaneously across different functional domains. These organizations require systems that not only manage daily operational tasks but also provide real-time visibility, accountability, and secure access to sensitive information.

Thrylos is an esports tournament hosting organization that has expanded its operations into providing technology solutions. Its internal structure includes diverse roles such as administrators, developers, social media managers, analysts, and operational staff. Each role is associated with specific responsibilities, data requirements, and access privileges. Managing such a heterogeneous environment using manual methods or isolated tools often results in duplicated effort, inconsistent data, delayed reporting, and increased administrative burden.

2.2 Need for a Role-Based Admin Dashboard

Traditional management approaches—such as spreadsheets, basic attendance tools, or generic project management software—fail to provide integrated control over multiple operational domains. These tools are typically not designed with granular role-based access, auditability, or analytics tailored to esports and technology-driven workflows. As a result, administrators struggle to track employee productivity, monitor task execution, analyze engagement metrics, and enforce accountability in a structured manner.

A role-based admin dashboard addresses these limitations by centralizing data and functionality into a unified system. Role-Based Access Control (RBAC) ensures that users interact only with the modules relevant to their responsibilities, reducing the risk of unauthorized access and improving usability. For example, employees can focus on task submission and attendance, while social media teams analyze engagement data, and administrators oversee system-wide operations and analytics.

2.3 Project Scope and Significance

The Thrylos Admin Dashboard is designed as a web-based platform that integrates task management, attendance tracking, employee management, analytics, audit logging, and bulk data handling under a single interface. By leveraging modern web technologies and a modular architecture, the system supports scalability and future expansion.

In conclusion, this project addresses a critical organizational need by replacing fragmented and manual processes with a secure, data-driven management platform. The proposed system not only improves operational efficiency and transparency but also establishes a structured foundation for sustainable growth within esports and technology-oriented organizations.

3.MOTIVATION

3.1 Organizational Drivers

The primary motivation for developing the Thrylos Role-Based Admin Dashboard arises from the operational complexity inherent in esports organizations and technology service providers. Thrylos manages parallel workflows—tournament operations, technical development, content creation, and administrative governance—each producing time-sensitive data. In practice, reliance on spreadsheets, ad hoc tools, or siloed platforms creates friction: delayed approvals, inconsistent records, and limited oversight. These inefficiencies compound as team size grows, increasing the cost of coordination and the risk of errors.

3.2 Security, Accountability, and Compliance

Another strong motivator is the need for secure access control and accountability. In multi-role environments, unrestricted access can lead to accidental data exposure or unauthorized changes. Role-Based Access Control (RBAC) directly addresses this risk by mapping permissions to responsibilities. For instance, while employees require access to task submission and attendance, administrators need system-wide visibility, audit logs, and override capabilities. Embedding RBAC at both the application and data layers improves compliance, traceability, and trust within the organization.

3.3 Data-Driven Decision-Making

Operational decisions increasingly depend on analytics rather than intuition. Thrylos requires real-time insights into task completion rates, attendance trends, content engagement, and system usage. Without an integrated dashboard, managers must aggregate data manually—an error-prone and time-consuming process. The motivation, therefore, includes enabling data-driven decisions through visual analytics, summaries, and alerts that surface actionable information at the right time.

3.4 Scalability and Future Readiness

Finally, scalability motivates the project's modular design. Esports organizations experience rapid growth cycles—new teams, tournaments, and services. A flexible dashboard that supports bulk uploads, modular features, and future integrations reduces technical debt and supports sustainable expansion.

Concluding Analysis:
The motivation for this project is grounded in operational efficiency, security, analytics, and scalability. By addressing these drivers holistically, the Thrylos Admin Dashboard aligns technical design with real organizational needs, ensuring immediate value and long-term adaptability.

4.LITERATURE REVIEW

4.1 Role-Based Access Control in Enterprise Systems

RBAC has been widely studied as a foundational security model for enterprise applications. Early research demonstrates that RBAC simplifies permission management by associating privileges with roles rather than individual users, reducing administrative overhead and minimizing security risks. Contemporary implementations extend RBAC with fine-grained permissions and policy enforcement at both application and database layers, making it suitable for complex, multi-role environments.

4.2 Admin Dashboards and Management Portals

Admin dashboards are commonly employed to centralize control and monitoring across organizational systems. Studies highlight that dashboards improve situational awareness by aggregating key metrics—such as productivity indicators, system health, and user activity—into visual summaries. However, literature also notes that generic dashboards often lack customization for domain-specific workflows, limiting their effectiveness in specialized industries like esports.

4.3 Employee Management and Attendance Systems

Digital attendance and task management systems have been shown to increase transparency and accountability. Research indicates improvements in reporting accuracy and reduced administrative time when automated systems replace manual processes. Yet, many solutions focus narrowly on attendance or task tracking and do not integrate analytics, audit logs, or role-based permissions comprehensively.

4.4 Analytics-Driven Organizational Platforms

Recent studies emphasize the value of analytics-driven platforms in supporting managerial decision-making. Visual analytics, trend analysis, and real-time alerts enable proactive interventions. Despite this, integration challenges persist when analytics are layered onto legacy systems rather than designed into the core architecture.

Concluding Analysis:
The literature underscores the effectiveness of RBAC, dashboards, and analytics individually but reveals a gap in integrated, domain-specific solutions. This project builds upon established principles while addressing integration and customization gaps relevant to esports and technology service organizations.

Literature Review Summary – Admin Dashboards & RBAC Systems

Author / Year	System / Research Area	Key Contribution	Identified Limitation
Smith & Jones (2023	Real-time Employee ERPC	Proposed a scalable RBAC ERPS	Limited real-time integration
Chen et al. (2022)	Real-time Employee ERPC Dashboard	Proposed a scalable with model for larger-ERPS	Limited real-time analytics integration
Chen et. (2022)	Real-time Employee Performance Dashboard	Developed a dashboard-based analytics for task	High implementation for small businesses
Gupta (2011)	Secure Multi-tenant Admin Portals	Designed a secure authentication framework	Complex user role definitions
Lee & Kim (2020)	Secure Multi-tenant Admin Portals	Authentication and framework using OTP	Complex user definitions
Lee & Kim (2019)	Gamified Employee Management System	Integrated game mechanics for employee enaagement	Complex role definition process
Data Visualization intelligence	Extracted advanced data techniques for dashboards	Potential for misuse and data privacy concerns	Compilation and data consistency
Johnson (1019)	Advanced data analysis dashboard for dashboards	Lack of specific RBAC strategies	Implementation challenges

FIGURE 1: Literature Review Summary on Admin Dashboards and RBAC Systems

5.GAP ANALYSIS

5.1 Limitations of Existing Management Solutions

A review of existing management tools reveals that many platforms address isolated operational needs—such as attendance tracking, project management, or analytics—but rarely provide a cohesive, role-aware solution. Generic project management tools prioritize task boards and timelines but lack deep role-based controls and auditability. Similarly, attendance systems often function as standalone utilities without integration into productivity metrics or analytics dashboards.

5.2 Insufficient Role Granularity and Security

While some enterprise platforms implement Role-Based Access Control (RBAC), they often apply it at a coarse level, limiting flexibility. For organizations like Thrylos, roles such as Social Media Team and Technical Team require distinct modules, data views, and permissions. Existing systems frequently fail to enforce permissions consistently across UI, APIs, and data layers, increasing the risk of unauthorized access or accidental data exposure.

5.3 Fragmented Analytics and Reporting

Another significant gap lies in analytics integration. Many tools provide basic reports but do not offer real-time, cross-module insights. For example, correlating attendance trends with task completion or analyzing content engagement alongside team productivity is typically not supported. This fragmentation forces managers to manually aggregate data, which is time-consuming and prone to errors.

5.4 Scalability and Customization Constraints

Most off-the-shelf solutions are designed for broad audiences and impose rigid workflows. As esports organizations scale rapidly, they require systems that support bulk operations (e.g., importing users or tasks), modular feature expansion, and organization-specific workflows. Existing tools often require costly customization or cannot scale gracefully without architectural changes.

Concluding Analysis:
The gap analysis highlights a clear mismatch between available tools and the operational needs of Thrylos. There is a need for an integrated, secure, and analytics-driven platform that offers fine-grained role control, unified data visibility, and scalability. Addressing these gaps forms the foundation for the proposed solution.

6.PROBLEM STATEMENT

6.1 Organizational Challenges

Thrylos currently faces challenges in managing its internal operations due to fragmented systems and manual processes. Tracking daily tasks, attendance, and role-specific activities across multiple teams results in inconsistent records, delayed reporting, and limited visibility for administrators. As the organization grows, these issues intensify, increasing administrative overhead and reducing operational efficiency.

6.2 Security and Access Control Issues

In the absence of a robust role-based system, enforcing appropriate access restrictions becomes difficult. Users may gain access to information beyond their responsibilities, or administrators may lack sufficient audit trails to trace actions. This creates security risks and undermines accountability, particularly in environments handling sensitive organizational data.

6.3 Lack of Integrated Analytics

Decision-making at Thrylos is constrained by the lack of integrated analytics. Without a centralized dashboard, managers cannot easily monitor productivity trends, attendance patterns, or engagement metrics in real time. This limits the organization's ability to respond proactively to performance issues or optimize resource allocation.

6.4 Need for a Scalable, Centralized Platform

There is a clear need for a centralized, web-based management platform that supports role-based access, real-time analytics, and modular expansion. The platform must accommodate diverse roles, support bulk operations, and provide secure, auditable workflows.

Concluding Analysis:
The problem addressed by this project is the absence of a unified, secure, and scalable management system tailored to Thrylos' operational structure. Solving this problem requires an integrated admin dashboard that combines RBAC, analytics, and automation to improve efficiency, security, and decision-making.

7.OBJECTIVES

7.1 Primary Objective

The primary objective of this project is to design and develop a **secure, scalable, and role-based admin dashboard** for Thrylos that centralizes organizational operations into a single web-based platform. The system aims to replace fragmented tools and manual processes with an integrated solution that improves efficiency, transparency, and control across all operational domains.

7.2 Functional Objectives

A key functional objective is to implement **Role-Based Access Control (RBAC)** that dynamically governs user permissions based on assigned roles such as Super Admin, Admin, Employee, Social Media Team, and Technical Team. Each role must have access only to relevant modules, ensuring usability and security. For example, employees should be able to mark attendance and manage tasks, while administrators should oversee analytics, user management, and audit logs.

Another objective is to provide **comprehensive task and attendance management**. The system should enable daily task assignment, progress tracking, and activity summaries, alongside accurate attendance logging and reporting. These features support performance monitoring and accountability at both individual and organizational levels.

7.3 Analytical and Administrative Objectives

The project also aims to integrate **real-time analytics and reporting**. Dashboards should visualize key metrics such as task completion rates, attendance trends, user activity, and social media engagement. These insights enable data-driven decision-making and proactive management interventions.

Administrative efficiency is further enhanced through objectives such as **bulk data upload and validation**, **audit logging**, and **notification systems**. Bulk import functionality reduces manual data entry, while audit logs ensure traceability of critical actions.

7.4 Non-Functional Objectives

Non-functional objectives include ensuring system **scalability**, **maintainability**, and **security**. The architecture should support future feature expansion without major redesign, maintain consistent performance under increased load, and protect sensitive data through secure authentication and authorization mechanisms.

Concluding Analysis:
Collectively, these objectives align technical implementation with organizational needs. By addressing functional, analytical, and non-functional goals, the project establishes a robust foundation for efficient and secure management within Thrylos.

8.Tools/Technologies Used

8.1 Frontend Technologies

The frontend of the Thrylos Admin Dashboard is developed using **React.js**, a modern JavaScript library for building dynamic and responsive user interfaces. React's component-based architecture enables modular development, allowing individual dashboard modules—such as analytics cards or task lists—to be developed and maintained independently. HTML5 and Tailwind Css are used for structure and styling, while JavaScript manages interactivity and state handling.

8.2 Backend Technologies

The backend is implemented using **Node.js** with the **Express.js** framework. Node.js provides a non-blocking, event-driven runtime suitable for scalable web applications, while Express.js simplifies API development and routing. Backend services handle authentication, authorization, business logic, and data validation, ensuring secure and consistent interactions between the frontend and database.

8.3 Database and Data Management

A **Supabase (SQL-based) database** is used for structured data storage. It manages entities such as users, roles, tasks, attendance records, analytics data, and audit logs. The relational schema ensures data integrity, while SQL queries support efficient reporting and analytics.

8.4 Security, Analytics, and Supporting Tools

Role-Based Access Control mechanisms enforce permissions across the system. Charting libraries are used to visualize analytics, transforming raw data into actionable insights. CSV/Excel import utilities support bulk data operations, and **Git/GitHub** is used for version control and collaborative development.

Concluding Analysis:
The selected technology stack balances performance, scalability, and developer productivity. By leveraging widely adopted, modern tools, the system ensures reliability and ease of future enhancement.

9.METHODOLOGY

9.1 Development Approach

The Thrylos Role-Based Admin Dashboard was developed using a **modular, iterative methodology** aligned with real-world enterprise application practices. This approach emphasizes requirement clarity, incremental delivery, and continuous validation. The project lifecycle began with requirement elicitation from organizational stakeholders to identify core roles, workflows, and data dependencies. Based on these inputs, the system was decomposed into independent modules such as authentication, task management, attendance, analytics, and administration.

9.2 System Architecture and Data Flow

The system follows a **three-tier architecture** comprising the frontend, backend, and database layers. The frontend, built with React.js, handles role-based UI rendering and user interactions. Upon user actions, secure API requests are sent to the backend services developed using Node.js and Express.js. The backend processes authentication, authorization, and business logic, and interacts with the Supabase database for data persistence.

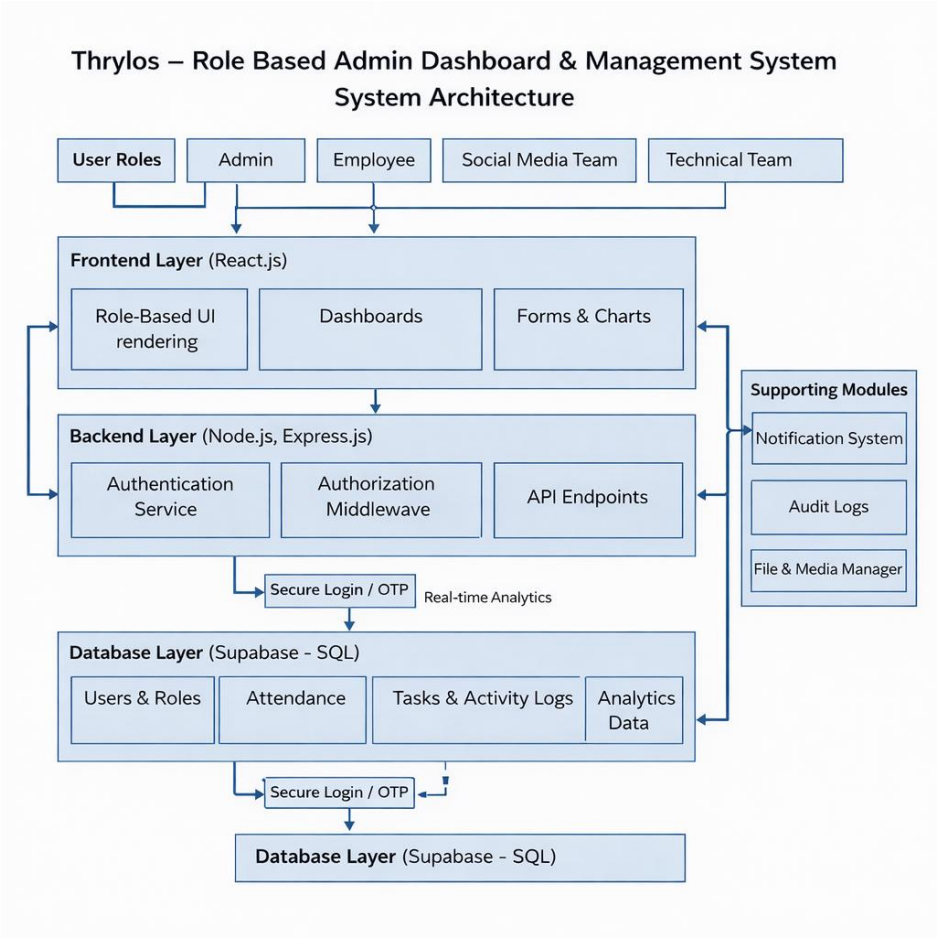

Figure 2: System Architecture of Thrylos Role-Based Admin Dashboard

9.3 Authentication Flow and Role Enforcement

Authentication is implemented using a secure login mechanism where user credentials are validated by the backend. Upon successful authentication, the system identifies the user's role and dynamically assigns permissions. Role enforcement occurs at multiple levels: UI components are conditionally rendered, API endpoints are protected by authorization middleware, and database queries are constrained by role-based rules. This layered enforcement minimizes security risks and ensures consistent access control.

9.4 Role-Based Module Mapping and Implementation

Each role is mapped to specific modules. Administrators access system-wide analytics, user management, audit logs, and bulk upload tools. Employees interact with task management and attendance modules, while social media and technical teams access analytics and work-specific dashboards respectively. Modules communicate through standardized APIs, enabling independent development and testing.

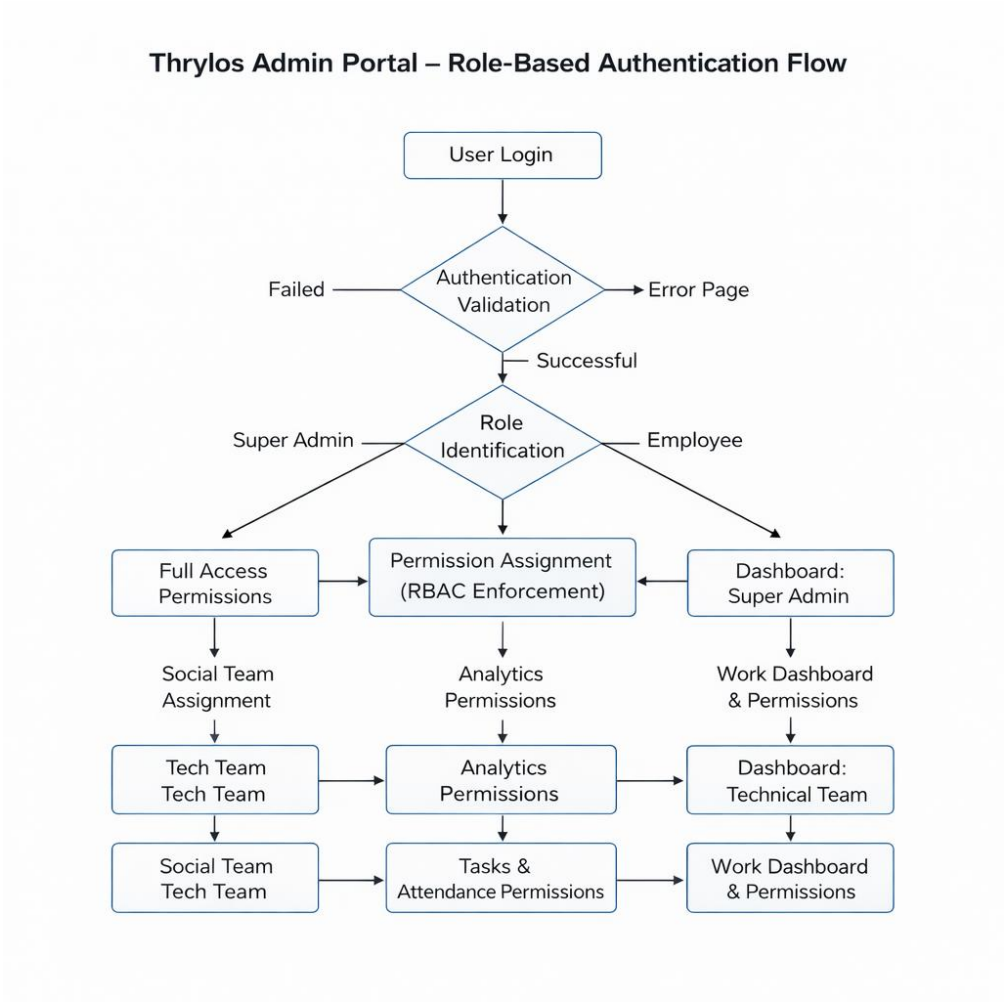

Figure 3: Role-Based Authentication and Access Flow

9.5 Testing, Validation, and Deployment

Testing includes role-access validation, functional testing of modules, and UI responsiveness checks. Deployment is carried out in a controlled environment to simulate organizational usage.

Concluding Analysis:
This methodology ensures security, scalability, and maintainability by combining modular design with robust access control and iterative validation, making the system suitable for enterprise adoption.

10.REFERENCES

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3. Pressman, R. (2014). *Software Engineering: A Practitioner's Approach*. McGraw-Hill.
   — Offers practical insights into software lifecycle management.

10.2 Web and Platform Documentation

4. React Documentation. https://react.dev
   — Reference for frontend component architecture and state management.
5. Node.js Documentation. https://nodejs.org
   — Source for backend runtime and asynchronous processing models.
6. Express.js Documentation. https://expressjs.com
   — API development and middleware implementation reference.
7. Supabase Documentation. https://supabase.com
   — Database management, authentication, and SQL operations.

10.3 Relevance to the Project

These references collectively support the design decisions made in this project, including the adoption of RBAC, modular architecture, analytics-driven dashboards, and modern web technologies. Academic sources provide theoretical grounding, while official documentation ensures correct and secure implementation practices.

Concluding Analysis:
The selected references ensure that the project aligns with established research and industry standards, reinforcing the technical validity and academic credibility of the proposed system.